

Repair Bias in Data

Project: Transparent Book Recommender

1. Your mission

Your learning target

I can identify selection and coverage bias.

I can compare groups before combining results.

I can explain why more biased data may not help.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Sampling and Representation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Noise and Outliers. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 20: Repair Bias in Data

Fairness work changes data, features, tests, or use conditions, then rechecks outcomes.



THINKER MISSION

Redesign an imbalanced speech dataset and specify retests.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

What is the difference between a population and a sample?

This week's end goal

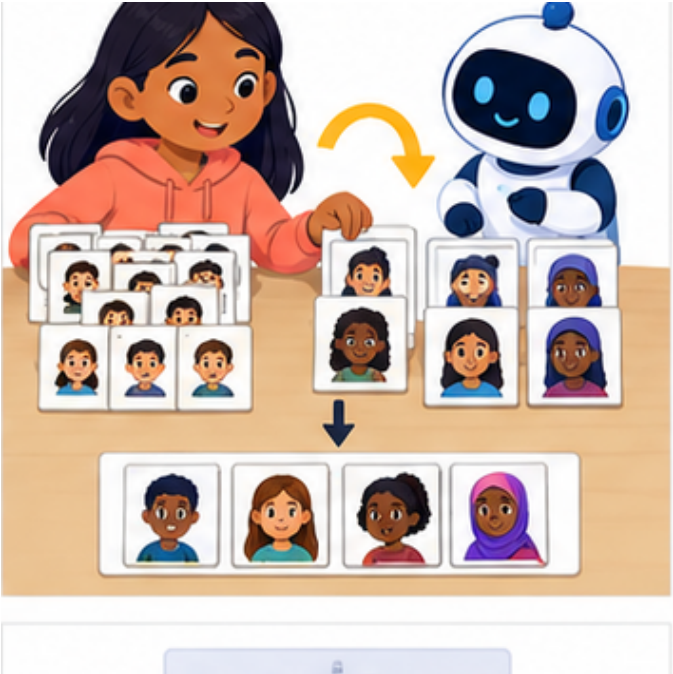
By the end, I can distinguish a biased collection process from a mere small dataset.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use repair bias in data to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

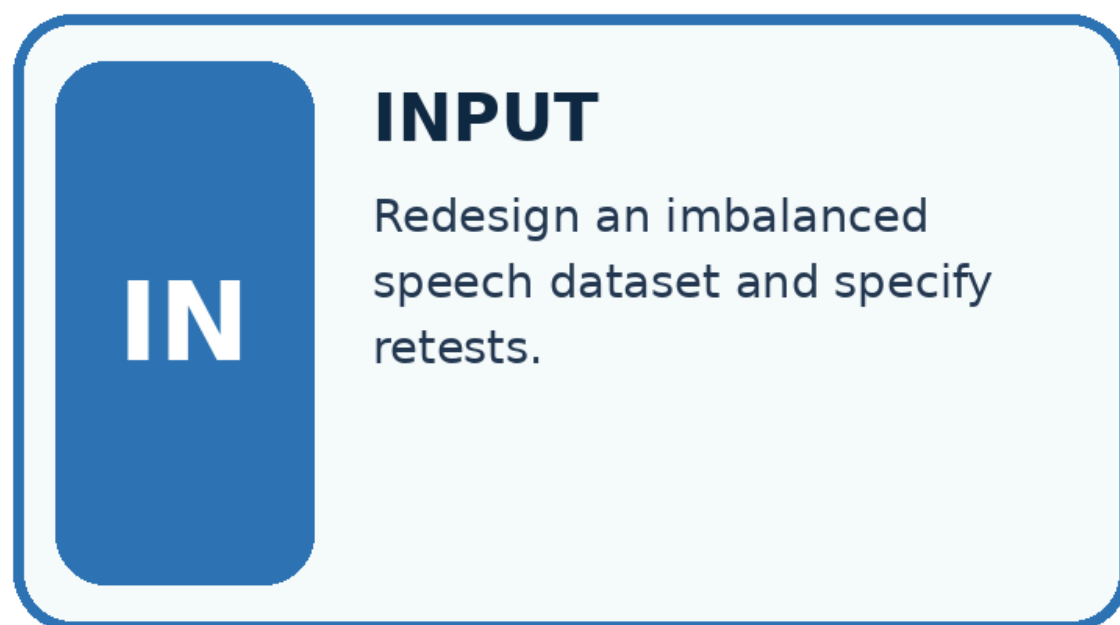
A fictional poll has 8 Yes among 10 club respondents and 2 Yes among 10 other respondents. If only club members are surveyed, an 80 percent result misses a group with a different pattern. Examine who is included before treating a larger sample as better.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



PROCESS

Fairness work changes data, features, tests, or use conditions, then rechecks outcomes.



OUT

OUTPUT

Redesign an imbalanced speech dataset and specify retests.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

```
club_yes, club_n = 8, 10
other_yes, other_n = 2, 10
print(club_yes / club_n)
print(other_yes / other_n)
print((club_yes + other_yes) / (club_n + other_n))
```

Follow the evidence

Step or case · Evidence · Meaning

Step or case: Club sample

Evidence: 8 of 10

Meaning: 80 percent

Step or case: Other sample

Evidence: 2 of 10

Meaning: 20 percent

Step or case: Combined sample

Evidence: 10 of 20

Meaning: 50 percent

Read each row from left to right and connect the evidence to the stated rule.

The combined observed sample has 50 percent Yes. This is not automatically a population estimate: we also need to know how the groups were selected and how many people each represents. Do not average percentages without considering their denominators.

Combine counts, not just percentages

The club has 8 Yes out of 10, and the other group has 2 Yes out of 10. Combine the Yes counts to get 10 and combine the response counts to get 20; the combined observed rate is 10 divided by 20, or 50 percent. When group sizes differ, simply averaging their percentages can give the wrong combined rate. A

correct combined calculation still describes the collected sample, not automatically the whole school.

Connect the picture and the explanation

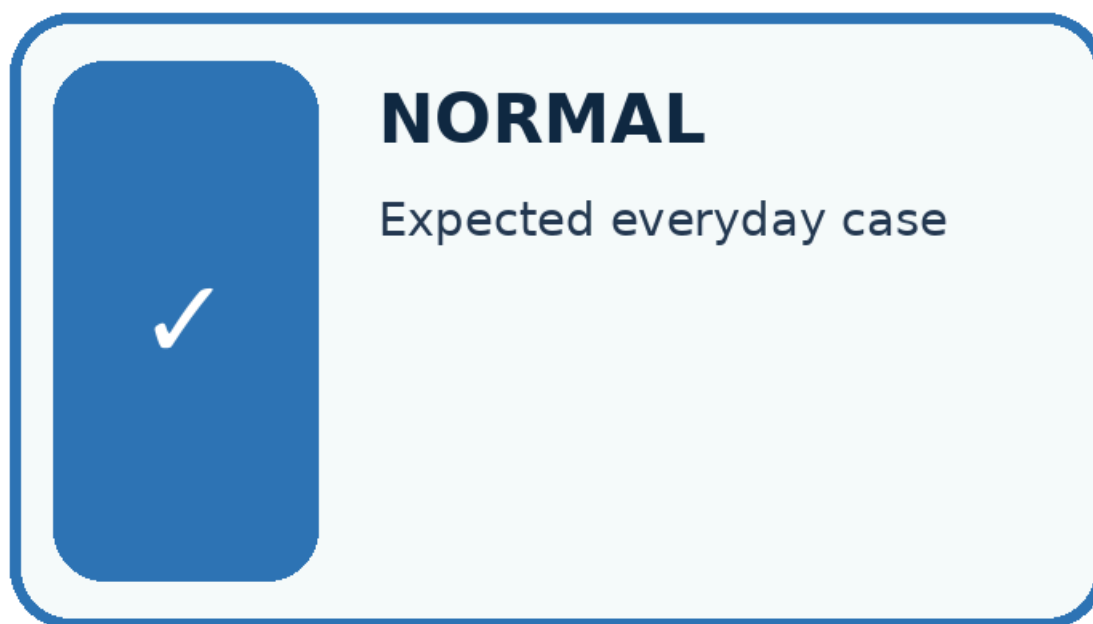
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

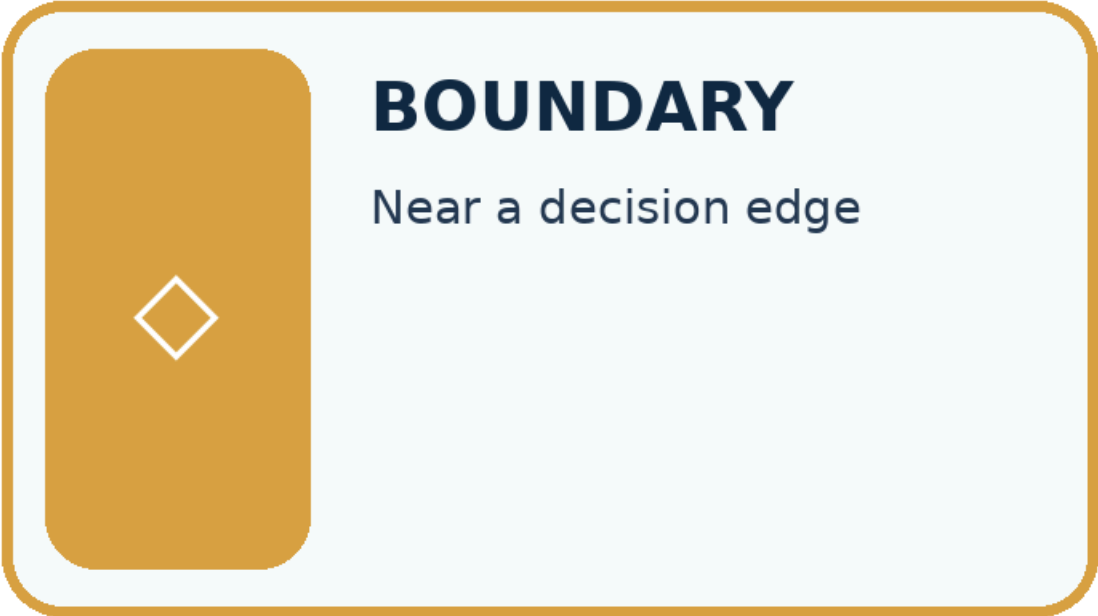
4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.





BOUNDARY

Near a decision edge

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING

Needed information absent



MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Recruit more club volunteers only.

Prediction: Sample size increases without fixing missing representation of other learners.

Reason: The added respondents come from the already overrepresented group.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

BIAS

systematic pattern of
uneven error or
disadvantage

2

REBALANCE

change coverage across
groups or conditions

3

SUBGROUP

meaningful part of the population

4

RETEST

repeat evaluation after a controlled change

Word · My own example · Not this / boundary

Word: bias

My own example: A systematic tendency in data, decisions or outcomes that can distort a result or disadvantage people.

Not this / boundary: _____

Word: rebalance

My own example: Change the relative representation of groups or classes in data using a stated method.

Not this / boundary: _____

Word: subgroup

My own example: A defined subset of cases considered separately in analysis.

Not this / boundary: _____

Word: retest

My own example: Run a test again after a change or under specified new conditions.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Selection bias

Meaning: Systematic distortion caused by who enters the data

Word: Undercoverage

Meaning: Leaving part of the target group out of selection

Word: Nonresponse

Meaning: A selected participant does not respond

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 20 • WORKED EVIDENCE

- 1 Audit: quiet-room voices work; noisy-room voices often fail.
- 2 Possible causes: missing environments, microphone differences, narrow speakers.
- 3 Repair: collect consented examples across settings; remove shortcut features.
- 4 Retest the same subgroup cases and report before/after results.

1. Audit: quiet-room voices work; noisy-room voices often fail.
2. Possible causes: missing environments, microphone differences, narrow speakers.
3. Repair: collect consented examples across settings; remove shortcut features.
4. Retest the same subgroup cases and report before/after results.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Suppose 90 club respondents give 72 Yes and 10 others give 2 Yes. What is the combined observed proportion?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Repair Bias in Data

A fictional school polls learners about adding comic recommendations.

The facts we will use

Library-club respondents: 12 Yes out of 15. Other respondents: 3 Yes out of 15.

The full grade has 30 club members and 90 others. These respondents were volunteers, not random selections.

AI ACADEMY / AI-2/20

Volunteer group comparison

	Yes responses	Respondents	Grade members
Club	12	15	30
Others	3	15	90

Read a cell using both its row and column labels.

Repair Bias in Data

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Group rates

Club: $12/15=80\%$; others: $3/15=20\%$.



2. Respondent total

15 Yes out of 30 respondents =50%.



3. Check representation

Respondents split 50/50, but the grade is 25% club and 75% other.



4. Repair collection

Plan including equality, documented invitations and report nonresponse by group.

CHANGE ONE THING

Recruit more club volunteers only.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Group rates

What happens and why: Club: $12/15=80\%$; others: $3/15=20\%$.

Step: 2. Respondent total

What happens and why: 15 Yes out of 30 respondents =50%.

Step: 3. Check representation

What happens and why: Respondents split 50/50, but the grade is 25% club and 75% other.

Step: 4. Repair collection

What happens and why: Plan including equality, documented invitations and report nonresponse by group.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

The combined respondent result gives club members more influence than their share of the grade. A weighted illustration using 25% and 75% gives $0.25 \times 80\% + 0.75 \times 20\% = 35\%$, but volunteers may differ within each group, so weighting alone does not establish the true grade preference. Improve the collection process rather than announcing the weighted number as fact.

What this example does not establish

Neither raw nor weighted volunteer rates prove the whole grade’s opinion; privacy and consent rules still apply.

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: Adding more total data automatically repairs bias. Strong move: locate evidence, change one cause, and retest.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Group differences suggest that coverage matters, but do not explain why choices differ. Small group

counts also make percentages unstable. Use neutral group labels in fictional data rather than personal or sensitive attributes.

Club members favor coding more often. Does club membership necessarily cause that preference?

8. Practise together

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET evidence_record PREDICT expected_result RUN one normal case and one boundary case
COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST
EXPLAIN result, limitation, and next safe step

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Surveying another 100 people from the same undercovered source may make the sample larger without fixing the selection problem. Improve access to missing groups and record nonresponse; do not simply duplicate existing rows.

Practise with less help: Combine counts

Use workbook W2 for the connected example “Selection bias and coverage”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Selection bias and coverage. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → TEST → RECORD → EXPLAIN

Given an uneven dataset, name the missing coverage, choose one repair, predict an outcome, and rerun a fixed subgroup test set.

Given an uneven dataset, name the missing coverage, choose one repair, predict an outcome, and rerun a fixed subgroup test set.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



MASTERY LADDER

1 Name it • 2 Trace it • 3 Apply it • 4 Challenge it • 5 Explain limits

- 1 • NAME: Define the concept without repeating the textbook.
 - 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
 - 3 • APPLY: Solve a different example independently.
 - 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
 - 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.
- My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



TRANSPARENT BOOK RECOMMENDER

Change feature coverage or weights only after locating which reader profiles are underserved.

THIS WEEK’S CHECKPOINT Change feature coverage or weights only after locating which reader profiles are underserved.

SYSTEM MAP Reader need → allowed inputs → transparent rule or score → ranked candidates → explanation → test record → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Broaden recommender feedback collection and show respondent group coverage before making user-preference claims.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT PROOF

Redesign an imbalanced speech dataset and specify retests.

CHALLENGE Redesign an imbalanced speech dataset and specify retests.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 20 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Training has eighteen adventure readers and two poetry readers; many intended users prefer poetry.

The question

Propose repair and evaluation.

Worked explanation

Collect varied relevant poetry examples with consistent labels. Compare both interests on separate evaluation cases before and after repair.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Repair coverage with varied examples

Evidence or stage · Value or result

Evidence or stage: Adventure readers

Value or result: 18

Evidence or stage: Poetry readers

Value or result: 2

Training has eighteen adventure readers and two poetry readers; many intended users prefer poetry.

Propose repair and evaluation.

Collect varied relevant poetry examples with consistent labels. Compare both interests on separate evaluation cases before and after repair.

14. Try a new case independently

Independent case: Audit independently

A survey has 18 Yes of 20 club members and 2 Yes of 10 others. Compute both group rates and combined rate. Explain why 20/30 does not automatically describe the whole grade.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 20 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A dataset has 18 adventure readers and 2 poetry readers. A recommender ignores poetry preferences. What would you inspect and test after a repair?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Change feature coverage or weights only after locating which reader profiles are underserved.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

A catalogue has 30 adventure titles and 2 poetry titles. A recommender rarely suggests poetry. Design a repair and an evaluation that distinguishes catalogue coverage from scoring preference.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.